W 11/L1 : Slide #26

$$f_{X,Y}(x,y) = \begin{cases} 2 & 0 \le y < x \le 1 \\ 0 & \text{O.W} \end{cases}$$

(1) Marginal PDF of Y:

slices of y goes from x to 1



$$f_Y(y) = \int_y^1 2\,dx \quad \text{for } 0 \le y < 1$$

$$= 2x\Big|_y^1$$

$$= 2 - 2y$$

$$f_Y(y) = \begin{cases} 2(1-y) & \text{for } 0 \le y < 1 \\ 0 & \text{O.W.} \end{cases}$$

(2) Conditional PDF of $Y|X$:

$$f_{X|Y}(x|y) = \frac{f_{X,Y}(x,y)}{f_Y(y)} \quad \text{for } y < x \le 1 \text{ w/ } 0 \le y \le 1$$

$$= \frac{2}{2(1-y)}$$

$$f_{X|Y}(x|y) = \begin{cases} \frac{1}{1-y} & \text{for } y < x \le 1 \\ 0 & \text{O.W.} \end{cases}$$

(3)